FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 4

Due: June 23, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

Important Note on Pace and Deadlines

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

The posted deadlines represent the maximum allowable completion time for each milestone and should not be interpreted as required pacing checkpoints. Early progress is strongly encouraged to allow additional time for refinement, testing, validation, and project improvement.

Milestone 4: Data Preprocessing & Baseline Modeling

Milestone 4 is designed to transition the capstone project from exploratory analysis to the preparation of a clean, modeling-ready dataset and the implementation of baseline machine learning or analytical models based on the **methodology presented in the selected base article for the capstone project**.

The preprocessing and baseline modeling process should clearly explain:

- How raw data was transformed into modeling-ready data,
- What preprocessing techniques were selected and why,
- How features were prepared and engineered,
- How training, validation, and testing datasets were created,
- Which baseline models were implemented,
- How baseline performance was evaluated,
- What limitations and opportunities were identified for future model improvement.

The primary objective of Milestone 4 is to implement and evaluate the reproducible baseline model identified in **Milestone 1**. This will serve as the foundation for comparing advanced modeling strategies, feature engineering improvements, and novel research contributions introduced throughout the remainder of the project.

1. **Data Preprocessing Implementation**

   Students must implement the preprocessing plan proposed in Milestone 3 and document all data preparation procedures.

   - **Preprocessing Requirements**

     Students must provide:
     - Complete preprocessing workflow,
     - Data cleaning procedures,
     - Missing value treatment methods,
     - Feature transformation procedures,
     - Data preparation decisions,
     - Justification for preprocessing choices,
     - Reproducible preprocessing methodology.
   - **Data Cleaning Activities**

     Students should explain any cleaning procedures performed, including:
     - Removal of duplicate records,
     - Correction of inconsistent values,
     - Standardization of formats,
     - Handling invalid or corrupted records,
     - Resolution of data integrity issues,
     - Dataset consolidation and verification.
   - **Missing Data Treatment**

     Students should document:
     - Missing value percentages,

  - Imputation techniques used,
  - Variables removed due to excessive missingness,
  - Impact of missing value treatment on dataset quality,
- **Outlier Treatment**

  Students should describe:
  - Outlier detection methods,
  - Outlier removal or transformation procedures,
  - Business rationale for retaining or removing extreme observations,
  - Impact on model readiness.

2. **Feature Engineering and Transformation**

   Students must prepare features suitable for machine learning or analytical modeling.

- **Feature Preparation Requirements**
  - Students should document:
  - Feature selection procedures,
  - Feature creation and engineering,
  - Variable transformations,
  - Data scaling and normalization,
  - Encoding methods for categorical variables.
- **Potential Techniques**

  Examples may include:
  - **Feature Engineering**
    - Derived variables,
    - Aggregated features,
    - Interaction variables,
    - Domain-specific engineered features,
    - Lag variables for time-series projects.
  - **Categorical Encoding**
    - One-hot encoding,
    - Label encoding,
    - Target encoding,
    - Frequency encoding.
  - **Feature Scaling**
    - Standardization,
    - Min-max normalization,
    - Robust scaling.
  - **Dimensionality Reduction (if applicable)**
    - Principal Component Analysis (PCA),
    - Feature extraction techniques,
    - Variable reduction methods.

3. **Data Partitioning Strategy**

   Students must establish appropriate datasets for model development and evaluation.

   - **Dataset Splitting Requirements**

     Students should clearly describe:

     - Training dataset,
     - Validation dataset,
     - Testing dataset,
     - Cross-validation strategy (if applicable).

   - **Discussion Topics**

     Students should explain:

     - Split ratios selected,
     - Randomization procedures,
     - Stratification methods (if applicable),
     - Time-based splitting approaches (for forecasting projects),
     - Measures taken to avoid data leakage.

4. **Baseline Model Development and Evaluation**

   Students are required to implement the baseline model(s) presented in the journal article selected during Milestone 1. The primary objective is to reproduce and validate the results reported in the source paper, thereby establishing a verified benchmark for subsequent experimentation and model enhancement activities in Milestone 5.

   Baseline Reproduction Requirements

   Students should:

   - Implement the baseline model architecture and methodology described in the selected journal article.
   - Recreate the experimental setup, including data preprocessing steps, model configurations, training procedures, and evaluation protocols, to the extent permitted by the information provided in the paper.
   - Document all implementation details, assumptions, and any deviations from the original methodology.
   - Compare the reproduced results with those reported in the source paper and discuss any similarities or discrepancies.
   - Evaluate model performance using the same or equivalent metrics reported in the paper.
   - Analyze factors that may influence reproducibility, such as dataset availability, implementation differences, parameter settings, or computational constraints.
   - Establish a validated baseline benchmark that will serve as the reference point for evaluating the effectiveness of proposed enhancements and novel contributions in Milestone 5.

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Milestone 4 Deliverables

Students must submit one professionally formatted document (PDF or Word) containing the following sections:

1. **Title Page**

   The title page should include:

   - Project Title
   - Student Group Name
   - Full Names of Team Members

2. **Updated Executive Summary (150–250 words)**

   The updated summary should briefly describe:

   - Data preprocessing activities,
   - Feature engineering efforts,
   - Baseline models developed,
   - Initial model performance,
   - Key predictive insights,
   - Major challenges encountered,
   - Planned optimization activities.

3. **Data Preprocessing Report**

   Students must include:

   - Data cleaning procedures,
   - Missing value handling methods,
   - Outlier treatment approaches,
   - Transformation techniques,
   - Data quality improvements,
   - Final modeling-ready dataset description.

4. **Feature Engineering and Data Preparation**

   Students must document:

   - Engineered features,
   - Encoding methods,
   - Scaling techniques,
   - Dimensionality reduction methods (if applicable),
   - Feature selection procedures,
   - Rationale for preprocessing decisions.

5. **Data Partitioning Strategy**

   Students must describe:

   - Training, validation, and testing datasets,
   - Dataset split methodology,
   - Cross-validation procedures,
   - Data leakage prevention measures.

6. **Baseline Model Development and Evaluation**

   This section must include:

   - **Model Description**
     - Selected algorithms,
     - Model assumptions,
     - Training procedures,
     - Configuration settings.
   - **Model Performance Results**

     Students must provide:
     - Performance metrics,
     - Evaluation tables,
     - Visualizations,
     - Comparative analysis of baseline models.
   - **Interpretation**

     Students should explain:
     - Model strengths and weaknesses,
     - Business relevance of results,
     - Opportunities for improvement.

7. **Baseline Modeling Assessment and Future Optimization Plan**

   Students must discuss:

   - Model limitations,
   - Improvement opportunities,
   - Planned advanced algorithms,
   - Hyperparameter tuning strategies,
   - Expected Milestone 5 activities.

8. **References (IEEE Style)**

   Students must use proper IEEE citation formatting consistently throughout the document.

   Example References

   [1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

   [2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

   [3] L. Johnson and M. Patel, "A hybrid NLP framework for social media sentiment analytics," *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics*, 2024.

Submission Guideline

Please submit one **Data Preprocessing & Baseline Modeling Report** per group through Canvas no later than **June 23, 2026, at 11:59 PM**.